

PESTS OF FIELD CROPS

Common name	Scientific name	Family	Order
Yellow stem borer	<i>Tryporyza/Scirpophaga incertulus</i>	Pyralidae	Lepidoptera
Rice hispa	<i>Diclodispa armigera</i>	Chrysomelidae	Coleoptera
Rice gundhi bug	<i>Leptocorisa acuta</i>	Coreidae	Hemiptera
Brown plant hopper	<i>Nilaparvata lugens</i>	Delphinidae	Hemiptera
Rice case worm	<i>Nymphula depunctalis</i>	Pyralidae	Lepidoptera

Rice crop pests

1. Yellow stem borer: *Tryporyza/Scirpophaga incertulus*.

- **Family:** Pyralidae (Lepidoptera)
- Monophagus/absolute pest of rice.
- Full grown larvae hibernate in rice stubbles.
- It pupates inside the plant.
- Full grown larvae measures about 20 mm adults are dirty white or greenish yellow front wings.

❖ **Symptoms of damage:**

- Caterpillar alone is destructive, construct an emergence hole which always located above the water level.
- Caterpillar produce dead hearts on affected plants.
- Plants that are attacked in early stages produce ears devoid of grains and are known as "white ear".
- Basmati varieties suffer heavy damage then coarse varieties.

❖ **Management:**

- Removal & destruction of crop stubbles.

- Since eggs of stem borer are laid near tip of leaf hence, clipping and burning of tips of the seedling as a routine before transplanting.
- Application of Cartap hydrochloride 4G @ 2.5 kg a.i. /ha.

2. **Rice gundhi bug:** *Leptocorisa acuta*

- Family: Coreidae
- Order: Hemiptera
- **Periodic pest of rice.**
- Adult bugs have long legs and are slender about 20 mm long.

❖ **Symptoms of damage:**

- Rice field severely attacked by this pest emit a repugnant smell which gives to this pest the name "Gundhi bug".
- The nymphs and adults suck juice from developing grains in the milky stage, causing incompletely filled panicles or panicles with empty grains.
- Black and brown spots are appearing around the holes made by the bugs on which a sooty mould may develop.
- This pest attacks the rice crop periodically.

❖ **Management:**

- Clean cultivation: Paddy field and surrounding bunds should be kept clean from weeds.
- Adult bugs captured through light traps and destroyed.
- Hand collection of bugs.
- **Resistant variety:** Grow Badshabhog variety.
- Spray Malathion 50 EC @ 675 ml/ha

3. **Brown plant hopper:** *Nilaparvata lugens*

- **Family:** Delphacidae
- Order: Hemiptera
- Serious pest of high yielding varieties of paddy.

❖ Symptoms:

- Both nymph & adult cause damage by sucking cell sap from leaves which turn yellow.
- Under the favorable condition of high humidity, high nitrogen application and no wind, the population increase and "**hopper burn**" is obtained in various localities.
- It transmitted the "grassy stunt" virus disease of rice.

❖ Management:

- Closer spacing should be avoided (follow a spacing of 20 x 15 cm).
- Alternate drying & wetting of field.
- Alley 30 cm wide after every 3 m of rice planting provide proper aeration to crop.
- Spray of 100 ml Imidacloprid (200 SL) with 250 liters of water.
- Multi gene resistant varieties: IR-36, IR-60.

❖ Other pests of rice:

a. Green leaf hopper: *Nephotettix nigropictus/virescens*.

- **Order:** Hemiptera
- **Family:** Cicadellidae
- This is responsible for vector of virus diseases, of which **Tungro** is most serious.

b. White backed plant hopper: *Sogatella furcifera*

- Family: Delphacidae
- Order: Hemiptera
- Cause rust red symptoms

c. Rice gall midge: *Orseolia oryzae*

- **Family:** Cecidomyiidae
- **Order:** Diptera
- Cause "silver shoot & onion leaves" symptoms.

d. Rice thrips: *Stenochaelothrips biformis*

- **Family:** Thripidae
- **Order:** Thysanoptera
- Important pest of rice nursery

e. Rice grasshopper: *Hieroglyphus banian*

- **Family:** Acrididae
- **Order:** Orthoptera
- Damage by nibbling of rice ear head

❖ **IPM Modules for Rice:**

- Hispa: 1-2 adult/hill
- BPH: 5-10 hopper/hill

❖ **Summary points for management**

- Avoiding use of excess nitrogen which increases population of Brown plant hopper (BPH) and leaf folder.
- Alternate wetting & drying for BPH.
- Avoid close planting in case of BPH epidemic area.
- Passing the rope on the crop and draining of water for case worm.
- Using the light traps to monitor BPH, Green plant hopper and stem borer.
- Clipping of seedling tips for rice hispa, thrips & yellow stem borer.
- Use of NSKE (Neem seed kernel extract) 50% or Neem oil for ear head bug

Insect -pest of wheat

Common name	Scientific name	Family	Order	Damaging stage
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Army worm	Mythimna separata	Noctuidae	Lepidoptera	Larvae
Ghujhia weevil	Tanymecus indicus	Curculionidae	Coleoptera	Adult

1. Army worm of wheat: *Mythimna separate*

- **Family:** Noctuidae
- **Order:** Lepidoptera.
- The freshly emerged larvae spin threads from which they suspend themselves in the air and then with the help of air current reach from one plant to another.
- Early-stage feeds on tender leaves in to central whorl of the plants but later caterpillars are able to feed on older leaves and skeletonized them totally.
- In case of severe attack by the army worm whole leaves including the mid ribs are consumed and the field looks as "grazed by the cattle".
- Spray Quinalphos 25 EC @ 500 ml/ha.

2. Ghujhia weevil: *Tanymecus indicus*,

- **Family:** Curculionidae
- **Order:** Coleoptera.
- It is specific pest of wheat nursery.
- Damage caused by adult weevil only.
- They cut the germinating seedlings at the ground level.
- Mainly damage is done in the months of October- November.

3. Other pests of wheat:

a. Termite: *Odontotermes obesus*,

- **Family:** Termitidae
- **Order:** Isoptera.

b. Wheat aphid: *Macrosiphum miscanthi*

- **Family:** Aphididae

- **Order:** Hemiptera.

Insect-pests of maize:

1. Maize/ Sorghum stem borer: *Chilo partellus*

- **Family:** Pyralidae
- **Order:** Lepidoptera.
- Most destructive pest of maize/sorghum.
- Caterpillars infect the crop usually one month after sowing.
- It causes pin holes on leaves or "dead hearts" in stem.
- The infestation may be noticed till harvesting.
- Larvae remains hibernate in maize stubbles.
- ❖ **Management:**
 - Destroy stubbles, weeds & alternate hosts.
 - Application of Phorate 10 G @ 20 kg/ha or Carbofuran 3G @ 25 kg/ha.
 - Release of Cotesia (Apanteles) spp.

Common name	Scientific name	Family	Order
Maize aphid	<i>Rhopalosiphum maydis</i>	Aphididae	Hemiptera
European corn borer. (It doesn't occur in India)	<i>Ostrinia nubilalis</i>	Pyralidae	Lepidoptera
Asian maize borer (present in some part of India)	<i>O. furnacalis</i>		
Pink borer It mainly feeds on maize, rice, and wheat. It causes 'dead heart' on central shoot in wheat.	<i>Sesamia inferens</i>	Noctuidae	Lepidoptera

Insect -Pest of cotton:

Common name	Scientific name	Family	Order
Cotton jassids	<i>Amrasca biguttula</i>	Cicadelidae	Hemiptera
Cotton whitefly	<i>Bemisia tabaci</i>	Aleyrodidae	Hemiptera

Pink boll worm	Pectinophora gossypiella	Gelechiidae	Lepidoptera
Spotted boll worm	Earias insulana vitella	Noctuidae	Lepidoptera

1. Cotton jassids: *Amrasca biguttula*

- **Family:** Cicadelidae
- **Order:** Hemiptera.
- Destructive pest of American cotton.
- Nymphs are wedge shaped and adults are winged form.
- ❖ **Symptoms:**
 - Adult and nymph cause injury to crop is due to loss of sap and probably also due to the injection of toxins.
 - Yellowing and downward rolling/cupping of leaves due to toxins.

2. Cotton whitefly: *Bemisia tabaci*

- **Family:** Aleyrodidae
- **Order:** Hemiptera
- The pale-yellow body is slightly dusted with a white waxy powder.
- They have two pairs of pure white wings and have prominent long hind wings.
- ❖ **Symptoms:**
 - **Damage done by two types.**
 - (a) The vitality of the plants is lowered through the loss of cell sap.
 - (b) The normal photosynthesis is interfered with due to the growth of a sooty mould on the honey dew excreted by the insects.
 - It **transmits** several viral diseases including the leaf curl disease of tobacco, vein curling disease of okra and leaf curl of sesame.
- ❖ **Management:**

- Both jassids and whiteflies can be effectively managed by spraying of Imidacloprid 200 SL @ 75 g a.i. /ha or Thiomethoxam @ 45 g a.i. /ha. I Spray of Dimethoate @ 1 ml/liter of water

3. Pink boll worm: *Pectinophora gossypiella*

- **Family:** Gelechiidae
- **Order:** Lepidoptera
- Most destructive pest of cotton.
- The caterpillars are pink in colour and found inside the flower buds, panicles and the bolls of cotton.

❖ Symptoms:

- Damage done by caterpillar in various ways there is excessive shedding of the fruiting bodies.
- "**Rosette flower**" is characteristic symptoms.
- The total shedding is caused by all the boll worms collectively one half may be due to the attack of pink bollworm.
- **Double seed formation:** The two adjoining seed are joining together within damaged boll by pink boll worm.

❖ Management:

- Growing of Bt. cotton varieties is the effective method for boll worm complex.
- Destruction of cotton sprouts, alternative host plants or burning of the plant debris.
- Deep ploughing (with furrow turning plough) done by the end of February.
- **Larval:** Parasitoid *Bracon greeni*, *Chelonus pectinophorae*.

4. Spotted boll worm: *Earias insulana* / *E. vitelli*

- **Family:** Noctuidae
- **Order:** Lepidoptera.
- It is serious pest of cotton and okra.

- A series of longitudinally black spots on the body with dull green colour and having tiny stout bristles are the main identification mark of spotted boll worm.

❖ Symptoms:

- Cause heavy shedding of fruiting bodies.
- '**Flare square**' (**Flare-up**) is characteristic symptoms.
- In the attacked bolls the lint is spoiled by larval feeding.
- The infected bolls open prematurely and produce poor lint resulting in lower market values.

❖ Other pests of cotton:

Common name	Scientific name	Family	Order
Dusky cotton bug	<i>Oxycarenus laetus</i>	Lycaenidae	Hemiptera
Red cotton bug Red cotton bug stains the lint in ginning factories when it crushed with cotton	<i>Dysdercus singulatus</i>	Pyrrhocoridae	Hemiptera
American boll worm	<i>Helicoverpa armigera</i>	Noctuidae	Lepidoptera
Cotton leaf roller	<i>Sylepta derogata</i>	Pyralidae	Lepidoptera

Insect – pests of Sugarcane

Common name	Scientific name	Family	Order
Top borer	<i>Tryporhyza/ Scirpophaga nivella</i>	Pyralidae	Lepidoptera
Sugarcane leafhopper	<i>Pyrilla perpusilla</i>	Lophopidae	Hemiptera.

1. Top borer: *Tryporhyza/Scirpophaga nivella*

- **Family:** Pyralidae
- **Order:** Lepidoptera.
- Young plants attacked by caterpillars show characteristic reddish streaks on the mid rib.

- They show a number of shoot holes in leaves which ultimately cause dead hearts.
- After cane formation it attacks the terminal portion of canes causing "bunchy tops".

❖ **Management:**

- Collect and destroy moths and egg clusters.
- Cut the attacked shoots at ground level from April to June.
- Application of Carbofuran 3G @ 25 kg/ha or Cartap hydrochloride 4G @ 25 kg/ha in July.

2. Sugar cane leafhopper: *Pyrilla perpusilla*

- **Family:** Lophopidae
- **Order:** Hemiptera.
- Jet like body posture.
- The leaf hoppers are very agile and jumps around in large numbers making a patient noise when a person walks through heavily infested field.
- Owing to the loss of cell sap the leaves turn pale yellow and shrivel up later.
- Leaf hoppers secrete a thick transparent liquid known as honey dew on which black mould is developed.
- Leaves acquire a sickly black appearance, and the attacked crop can be spoiled from a distance.

❖ **Management:**

- Spray Malathion 50 EC @ 1 liter/ha.
- **Egg parasitoid:** *Epiricania melanoleuca* (single parasitoid belonging to order Lepidoptera).

Common name	Scientific name	Family	Order
Sugar cane white fly	<i>Aleurobolus barodensis</i>	Aleyrodidae	Hemiptera
Sugar cane mealy bug	<i>Saccharicoccus sacchari</i>	Pseudococcidae	Hemiptera